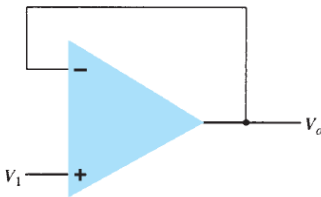
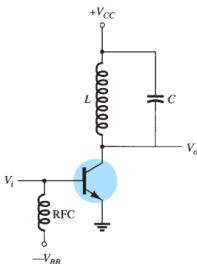
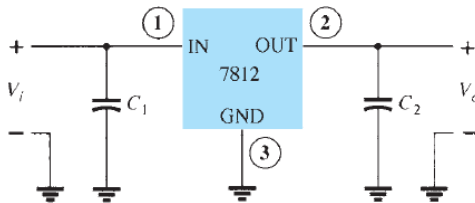
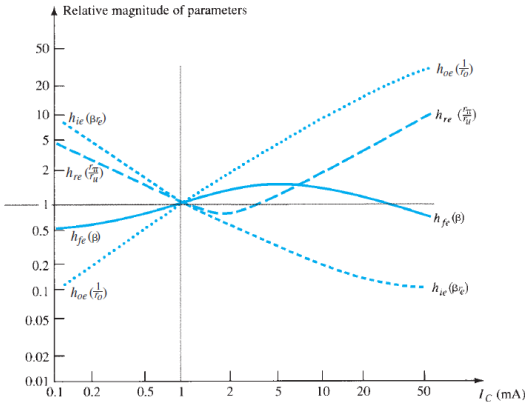
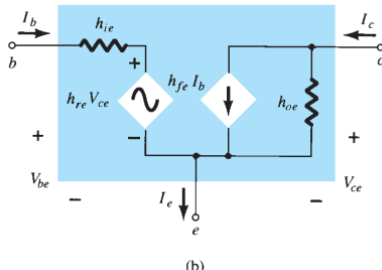
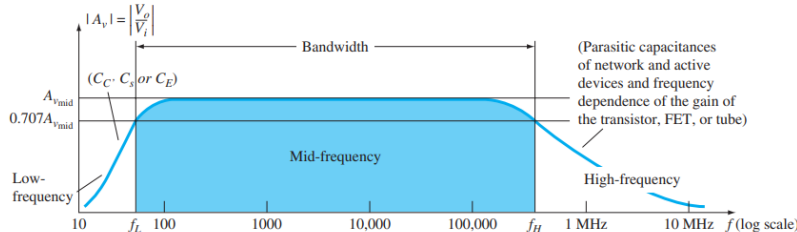
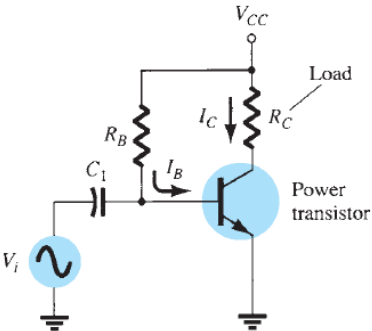


			Total Pages: 5
Answer Key			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
THIRD SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2021			
Course Code: BMT205			
Course Name: ANALOG ELECTRONICS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions. Each question carries 3 marks	Marks
1	It is defined for a set of operating conditions that might not match the actual operating conditions.		(3)
2	$I_E = I_C + I_B$ (1 mark) $\frac{I_C}{\alpha} = I_C + \frac{I_C}{\beta}$ (1 mark) $\alpha = \frac{\beta}{\beta+1}$ (1 mark)		(3)
3	- The unbypassed emitter resistance will have larger input impedance compared to the bypasses emitter resistance for CE emitter bias or votage divider bias configuration. (1 mark) - Bypassed: $Z_i = R' \beta r_e, R' = \frac{R_1 R_2}{R_1 + R_2}$ (1 mark) - Unbypassed: $Z_i = R' \beta (R_E + r_e), R' = \frac{R_1 R_2}{R_1 + R_2}$ (1 mark)		(3)
4	The only difference from the depletion-type MOSFET construction is the absence of the insulator at the gate. (1.5 marks) GaAs (1.5 marks)		(3)
5	$CMRR = \frac{A_d}{A_c}$ (1 mark) When separate inputs are applied to the op-amp, the resulting difference signal is the difference between the two inputs. $V_d = V_{i_1} - V_{i_2}$ $A_d = \frac{V_o}{V_d}$ (1 mark) When both input signals are the same, a common signal element due to the two inputs can be defined as the average of the sum of the two signals. $V_c = \frac{1}{2}(V_{i_1} + V_{i_2})$ $A_c = \frac{V_o}{V_c}$ (1 mark)		(3)
6	A voltage buffer circuit provides a means of isolating an input signal from a load by using a stage having unity voltage gain, with no phase or polarity inversion, and acting as an ideal circuit with very high input impedance and low output impedance. (1 mark)		(3)

		
7	The question is not correct because, we can setup a precision rectifier using op-amp but not precision diode. Hence marks needs to be awarded to students who attempted as precision rectifier.	(3)
8	 Circuit diagram (3 marks)	(3)
9	1. Higher input impedance. 2. Better stabilized voltage gain. 3. Improved frequency response. 4. Lower output impedance. 5. Reduced noise. 6. More linear operation. (Any three, 1 mark for each)	(3)
10		(3)
PART B		
<i>Answer any one full question from each module. Each question carries 14 marks</i>		
Module 1		
11	a. $I_{BQ} = 21.42 \text{ mA}$ (1 mark) $I_{CQ} = 1.71 \text{ mA}$ (1 mark) $V_{CEQ} = 8.17 \text{ V}$ (1 mark) $V_C = 9.33 \text{ V}$ (1 mark) $V_E = 1.16 \text{ V}$ (2 mark) $V_B = 1.86 \text{ V}$ (1 mark) b. The parameter $h_{fe}(\beta)$ varies the least of all the equivalent transistor circuit parameters when plotted against variations in collector current. (1 mark)	(7) (7)

		• $h_{oe}(1/r_o)$ and $h_{ie}(\beta r_e)$ vary the most for the chosen current range. (1 mark) • The parameter h_{ie} varies from about 10 times its Q-point value down to about one-tenth the Q point value at 50 mA. However, this variation should be expected because r_e is directly related to the emitter current by $r_e = 26 \text{ mV}/I_E$. (1 mark)	
		 (4 marks)	
12	a.	 (3 marks) $h_{ie} = \beta r_e$ $h_{fe} = \beta_{ac}$ $h_{ib} = r_e$ $h_{fb} = -\alpha \cong -1$ (Definition 6 marks)	(9)
	b.	The concept of thermal runaway is not mentioned in the syllabus	(5)
Module 2			
13	a.		(5)

	 (3 marks)	
	Explanation (2 marks) b. Low frequency response limited by 3 poles is beyond the syllabus. Because (9) the students haven't introduced the concept of poles and zeros. Deriving the cutoff-frequency due to C_C, C_E and C_S (7.5 marks) Definition of the overall lower cut-off frequency from the affect of C_C, C_E and C_S (1.5 marks)	
14	a. Output/transfer characteristics of enhancement type MOSFET and depetion type MOSFET (6 marks) n-channel enhancement type MOSFET (1.5 marks) $I_D = k(V_{GS} - V_{GS(Th)})^2$ $k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{GS(Th)})^2}$ n-channel depetion type MOSFET (1.5 marks) $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$ b. Short description on CMOS (1 marks) Features (4 marks)	(9)
Module 3		
15	a. Listing all the four basic blocks of an operational amplifier (1 mark) Explanation on each block-Differential amplifier (4 marks), other blocks (3.5 marks) (8 marks) b. Explanation (5 marks)	(9)
16	a. Circuit diagram of non-inverting amplifier (2 marks), designing the values of R_1 and R_f for gain =10 (3 marks) b. Instrumentation amplifier is not mentioned in the syllabus	(5)
Module 4		

17	a. Current to voltage converter is not mentioned in the syllabus. (7) b. Pin diagram (2 marks) Functionl diagram (3 marks) Explanation (2 marks) (7)	
18	 (3 marks) DC bias operation (3.5 marks) AC operation (3.5 marks) The maximum efficiency of a class A series-fed amplifier is thus seen to be 25%. (Derivation – 4 marks)	(14)
Module 5		
19	Colpitts Oscillator is not mentioned in the syllabus.	(14)
20	Different types of negative feedback: Voltage-Series Feedback, Voltage-Shunt Feedback, Current-Series Feedback, Current-shunt Feedback (8 marks) Derivation of the overall gain of any type of feedback (4 marks) Differentiating positive and negative feedback (2 marks)	(14)